

M2 Qbank (For CV/ME/AE) April/May 2026 Exams

MODULE 1: Numerical Methods

Section A

1. Construct a forward difference table for the data in given table:

x	10	11	12	13
y	22	24	28	34

- Write down the Newton's backward interpolation formula
- Write the Newton's general interpolation formula up to fourth divided differences.
- Write down the formula to find a real root of an equation $f(x) = 0$ using Regula-Falsi method.
- Find the initial approximation of the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 1$ using modified Euler's method taking $h = 0.05$

Section B

1. Using Lagrange's interpolation formula, find the value of y for $x=10$ given

x	5	6	9	11
y	12	13	14	16

- Employ Taylor's series method to find y at $x = 0.1$ correct to 4 decimal places considering up to fourth degree term given the equation $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$
- Use Newton-Raphson method to find a root correct to three decimals of the equation $3x = \cos x + 1$ near $x = 0.5$
- Use Weddle's rule to evaluate $\int_0^1 \frac{x dx}{1+x^2}$ taking seven ordinates and hence find $\log_e 2$

Section C

1. Evaluate $f(9)$ and $f(18)$ using Newton's divided difference formula, given

x	5	7	11	13	17
f(x)	150	392	1452	2366	5202

2. From the data given below, find the number of students who obtained (i) less than 45 marks and (ii) between 40 and 45 marks

Marks	30-40	40-50	50-60	60-70	70-80
No. of students	31	42	51	35	31

3. Apply Runge-Kutta method of fourth order to find an approximate value of y at $x = 0.1$ given that $\frac{dy}{dx} = x + y^2$ and $y = 1$ when $x = 0$

MODULE 2: PDE

Section A

- Form a partial differential equation from $z = (x + a)(y + b)$
- Form a partial differential equation from $z = f(x + ay)$
- Write down the partial differential equation for (i) one dimensional wave equation (ii) one dimensional heat equation

4. Write down the auxiliary equation for the Lagrange's P.D.E.: $(y + z)p + x(yq - z) = 0$
5. Form a partial differential equation from $z = ax + by + ab$

Section B

1. Form a partial differential equation by eliminating arbitrary constants from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
2. Solve $y^2 zp = x^2(zq + y)$
3. Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos(2x + 3y)$
4. Form a partial differential equation by eliminating the arbitrary functions f and g from $z = f(x + ay) + g(x - ay)$

Section C

1. Form a partial differential equation from $\phi(x + y + z, x^2 + y^2 + z^2) = 0$
2. Solve $\frac{\partial^2 z}{\partial y^2} = z$, given that when $y = 0$, $z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$
3. Obtain the various possible solutions of one dimensional heat equation by method of separation of variables

MODULE 3: ODE

Section A

1. Find the general solution of $(D^2 - 4)y = 0$
2. Find the particular integral of $(D^2 - 7D + 12)y = e^{2x}$ if the complementary function is $Ae^{3x} + Be^{4x}$
3. Find the complementary function of $(D^2 - 4D + 4)y = x^2$
4. Using the method of variation of parameters, find the Wronskian W , where $y_1 = \cos ax$ and $y_2 = \sin ax$
5. Write down the particular integral of $(D^2 + 9)y = \cos 3x$

Section B

1. Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 13y = \cos 2x$
2. Solve by inverse differential operator method $(D^3 - 5D^2 + 8D - 4)y = e^{2x}$
3. Solve by inverse differential operator method $(D^2 + D)y = x^2 + 2x + 4$
4. Solve $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 4y = e^x \cos x$

Section C

1. Solve by inverse differential operator method $(D^2 + 3D + 2)y = e^{2x} + (2x^2 + 4x + 1)$
2. Solve by inverse differential operator method $(D^2 - 4D + 4)y = e^{2x} + \cos 2x + x^2$
3. By the method of variation of parameters, solve $y'' + a^2 y = \sec ax$

MODULE 4: Laplace Transform

Section A

1. Show that $L[e^{at}] = \frac{1}{s-a}$

2. State the division by "t" property, of Laplace transform
3. Find the inverse Laplace transform of $\frac{3s^2+4}{s^5}$
4. Write the formula for shifting property of Heaviside shift theorem
5. Write the formula for the Laplace transform of a periodic function

Section B

1. Find the Laplace transform of $t^5 \cosh 3t$
2. Express $f(t) = \begin{cases} t^2, & 0 < t \leq 2 \\ 4t, & t > 2 \end{cases}$ in terms of unit step function and hence evaluate it.
3. Find the Laplace transform of the periodic function $f(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4 \end{cases}$
4. Evaluate using Laplace transform $\int_0^\infty t e^{-2t} \sin t \, dt$

Section C

1. Find $L \left[\left(\sqrt{t} + \frac{1}{\sqrt{t}} \right)^3 \right]$
2. Find the inverse Laplace transform of $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$
3. Solve by Laplace transform method: $y'' + 4y' + 3y = e^{-t}, y(0) = y'(0) = 1$